Likewise the statement says: $\forall x \exists y (x+y=0)$.
For all real numbers, there exists another real number such that their sum is 0.
Similarly, the statement $\forall x \forall y \forall z (x+(y+z) = (x+y)+z)$ is the associative law for the addition of real numbers.

Example 2: Translate into English the statement $\forall x \forall y ((x>0) \land (y<0) \rightarrow (xy<0))$, where the domain for both variables consist of all real numbers.

Soln: For any two arbitrary real numbers where one is positive and the other is negative, the product must be a negative real no.

1.5.3 The Order of Quantifiers

Example 3: Let $P(x,y)$ be the statement "$x+y=y+x$". What are the truth values of the quantifications $\forall x \forall y P(x,y)$ and $\forall y \forall x P(x,y)$, where the domain for all variables consists of all real numbers?

Soln: The quantification $\forall x \forall y P(x,y)$ denotes the proposition "For all real numbers x, for all real numbers y, $x+y=y+x$."
Because $P(x,y)$ is true for all real numbers x and y (it is the commutative law for addition, which is an axiom for the real numbers), the proposition $\forall x \forall y P(x,y)$ is true. Note that the statement $\forall y \forall x P(x,y)$ says "For all real numbers y, for all real numbers x, $x+y=y+x$." This has the same meaning as the statement "For all real numbers x, for all real numbers y, $x+y=y+x$." That is, $\forall x \forall y P(x,y)$ and $\forall y \forall x P(x,y)$ have the same meaning, and both are true.

This illustrates the principle that the order of nested universal quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement.